

HRN-v2: A Hierarchical Resonant Network for Sequential MNIST under Strict Biological-Plausibility Constraints

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Abstract

We present **HRN-v2**, a hierarchical resonant neural network whose every mechanism — forward dynamics, credit assignment, plasticity rule, and readout — is local in time and local in space. The substrate is a population of complex damped-rotation oscillators (one ODE per neuron, no recurrent inter-neuron coupling within a stage); the credit signal is a forward-only eligibility trace; the readout is a class-pool tail-energy projection; supervision is DECOLLE-style per-stage local cross-entropy. There is no backpropagation through time, no surrogate-gradient spike emission, and no symmetric backward pathway. Stages are stacked with a sparse class-aligned fan-in and a strict frequency-band hierarchy (faster $\rightarrow$ slower across stages). On Sequential MNIST (SMNIST, 784-step single-channel pixel sequence, 10-way classification, 10k training subset) the 3-stage HRN-v2 reaches **89.9%** test accuracy, more than 50 percentage points above the `pub_v3` single-layer self-organising baseline on the same task, and at the threshold of the 91–92% target the authors set as the biological-plausibility ceiling for this class of models.

1 Introduction and motivation

We adopt the working hypothesis that biological neural computation is naturally expressed in the language of *resonance*:

- *computation* is the act of an internal oscillator phase-locking with structure in its driving signal;
- *learning* is the local tuning of dynamical-system parameters (frequency, damping, input weights) of each oscillator;
- *memory* is the stable resonant configuration the system settles into for a class of inputs;
- *decision-making* is the act of one resonant mode dominating another at readout time.

A natural consequence of this view is *recursive decomposition*: a hard problem is broken into modes; each mode is handled by a pool of neurons; each pool itself decomposes its sub-problem into smaller modes at a slower time-scale. The architecture of HRN-v2 is built directly on this principle.

The substrate is a strictly faithful version of the linear damped complex oscillator from [1] (the per-neuron model used in `resonant_self_organizing_layer.py`); we deliberately remove the cubic Stuart–Landau term, the spike emission with surrogate gradient, and the inter-neuron recurrence used in earlier experiments [2], all of which are sources of either non-locality, non-stability, or pseudo-biological backpropagation.

2 Notation and assumptions

We index time by $t \in \{0, 1, \dots, T-1\}$ with $T = 784$ for SMNIST. The input at time t is a real vector $x_t \in \mathbb{R}^F$; for stage 0 with raw SMNIST input, $F = 1$ and x_t is the greyscale value of the t -th pixel in row-major raster order.

Each stage $\ell \in \{0, 1, 2\}$ contains P_ℓ oscillators. State of oscillator i at stage ℓ at time t is a complex scalar $z_{\ell,i}(t) = z_{r\ell,i}(t) + i z_{i\ell,i}(t) \in \mathbb{C}$, with amplitude squared $E_{\ell,i}(t) = |z_{\ell,i}(t)|^2 = z_{r\ell,i}^2(t) + z_{i\ell,i}^2(t)$.

Throughout we make the following assumptions:

- A1 (locality of plasticity).** Every parameter update of oscillator i at stage ℓ depends only on quantities (i) computable from the trajectory of i itself, (ii) the layer-local class-pool target signal, and (iii) homeostatic statistics of i . No update uses the gradient of another oscillator's loss.
- A2 (forward-only credit).** The plasticity rule does not require a backward pass. All credit signals are computed by forward eligibility traces evolved *simultaneously* with the state.
- A3 (no spike surrogates).** The output of each oscillator at the readout boundary is its tail-window mean amplitude squared $\bar{E}_{\ell,i} = \frac{1}{T_\ell} \sum_{t=T-T_\ell}^{T-1} E_{\ell,i}(t)$, where T_ℓ is the tail length for stage ℓ . There is no spike-emission step requiring a surrogate gradient.
- A4 (no inter-stage backward).** Stage ℓ does *not* receive a gradient from stage $\ell + 1$. Each stage has its own class-pool readout and its own cross-entropy loss; the only cross-stage signal is the forward feed of $E_{\ell,i}(t)$ into stage $\ell + 1$.
- A5 (strict frequency hierarchy).** The intrinsic frequency band for stage ℓ satisfies $[\omega_\ell^{\min}, \omega_\ell^{\max}] \supsetneq [\omega_{\ell+1}^{\min}, \omega_{\ell+1}^{\max}]$ with $\omega_{\ell+1}^{\max} \leq \omega_\ell^{\min} + \frac{1}{2}(\omega_\ell^{\max} - \omega_\ell^{\min})$, so each higher stage operates strictly slower than the stage below.

3 Per-neuron oscillator

A single oscillator with parameters (d, b, ω, α) where $d, b \in \mathbb{C}$, $\omega \in \mathbb{R}^+$, $\alpha \in (0, 1)$, evolves under the discrete-time *linear damped complex rotation*:

$$z(t+1) = \alpha e^{i\omega} z(t) + d \cdot x(t) + b, \quad z(0) = 0. \quad (1)$$

Real-imaginary form, vectorised across F input channels:

$$z_r(t+1) = \alpha \cos(\omega) z_r(t) - \alpha \sin(\omega) z_i(t) + \sum_{k=1}^F d_{rk} x_k(t) + b_r, \quad (2)$$

$$z_i(t+1) = \alpha \sin(\omega) z_r(t) + \alpha \cos(\omega) z_i(t) + \sum_{k=1}^F d_{ik} x_k(t) + b_i. \quad (3)$$

The intrinsic frequency ω and damping α are stored in unconstrained parameters $\omega^{\text{raw}}, \alpha^{\text{raw}} \in \mathbb{R}$ and mapped through sigmoids to controlled ranges:

$$\omega = \omega_{\min} + (\omega_{\max} - \omega_{\min}) \sigma(\omega^{\text{raw}}), \quad (4)$$

$$\alpha = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \sigma(\alpha^{\text{raw}}), \quad (5)$$

with $\sigma(u) = 1/(1 + e^{-u})$. We write $\omega'(\omega^{\text{raw}}) = (\omega_{\text{max}} - \omega_{\text{min}}) \sigma(1 - \sigma)$ for the chain-rule factor, and analogously $\alpha'(\alpha^{\text{raw}})$.

The *tail-energy* readout over the last T_{tail} steps is

$$\bar{E} = \frac{1}{T_{\text{tail}}} \sum_{t=T-T_{\text{tail}}}^{T-1} [z_r^2(t) + z_i^2(t)]. \quad (6)$$

This is the only quantity passed to the loss for a given neuron.

4 Forward eligibility traces

For each parameter $\theta \in \{d_{rk}, d_{ik}, b_r, b_i, \omega^{\text{raw}}, \alpha^{\text{raw}}\}$ and each oscillator we maintain a complex eligibility trace

$$e^\theta(t) = \begin{pmatrix} e_r^\theta(t) \\ e_i^\theta(t) \end{pmatrix} = \frac{\partial z(t)}{\partial \theta} = \begin{pmatrix} \partial z_r(t) / \partial \theta \\ \partial z_i(t) / \partial \theta \end{pmatrix}. \quad (7)$$

Differentiating (2)–(3) gives the recurrence

$$e^\theta(t+1) = R_\alpha(\omega) e^\theta(t) + \frac{\partial}{\partial \theta} [d \cdot x(t) + b] + \frac{\partial R_\alpha(\omega)}{\partial \theta} z(t), \quad (8)$$

where the 2×2 rotation+damping matrix is

$$R_\alpha(\omega) = \alpha \begin{pmatrix} \cos \omega & -\sin \omega \\ \sin \omega & \cos \omega \end{pmatrix}. \quad (9)$$

For input weights and biases the second term is a direct injection and the third term vanishes. For the k -th channel of d_r and d_i :

$$e^{d_{rk}}(t+1) = R_\alpha(\omega) e^{d_{rk}}(t) + \begin{pmatrix} x_k(t) \\ 0 \end{pmatrix}, \quad e^{d_{ik}}(t+1) = R_\alpha(\omega) e^{d_{ik}}(t) + \begin{pmatrix} 0 \\ x_k(t) \end{pmatrix}. \quad (10)$$

For the biases:

$$e^{b_r}(t+1) = R_\alpha(\omega) e^{b_r}(t) + \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e^{b_i}(t+1) = R_\alpha(\omega) e^{b_i}(t) + \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (11)$$

For ω^{raw} and α^{raw} , only the third term in (8) is non-zero. Define $\rho(t) = R_\alpha(\omega) z(t) / \alpha = \begin{pmatrix} \cos \omega z_r - \sin \omega z_i \\ \sin \omega z_r + \cos \omega z_i \end{pmatrix}$, the rotated state *before* damping. Then

$$\frac{\partial R_\alpha(\omega)}{\partial \omega^{\text{raw}}} z(t) = \alpha \omega'(\omega^{\text{raw}}) \begin{pmatrix} -\rho_i(t) \\ \rho_r(t) \end{pmatrix}, \quad (12)$$

$$\frac{\partial R_\alpha(\omega)}{\partial \alpha^{\text{raw}}} z(t) = \alpha'(\alpha^{\text{raw}}) \begin{pmatrix} \rho_r(t) \\ \rho_i(t) \end{pmatrix}, \quad (13)$$

giving the dedicated recurrences for the dynamic parameters.

4.1 Tail-energy gradient via eligibility

The tail-energy gradient is then a linear functional of the eligibility:

$$\frac{\partial \bar{E}}{\partial \theta} = \frac{2}{T_{\text{tail}}} \sum_{t=T-T_{\text{tail}}}^{T-1} \left[z_r(t) e_r^\theta(t) + z_i(t) e_i^\theta(t) \right], \quad (14)$$

which is the bio-eligibility identity for the squared amplitude.

Crucially, the only quantity that enters $\partial \bar{E} / \partial \theta$ at time t is the local pair $(z(t), e^\theta(t))$ — both stored at the post-synaptic neuron, both updated by (2) and (8) without involving any other neuron.

5 Stage-level architecture

A stage ℓ contains P_ℓ oscillators arranged into 10 *class-pools* of M_ℓ oscillators each ($P_\ell = 10 M_\ell$). The class index of oscillator i in stage ℓ is $c_\ell(i) \in \{0, \dots, 9\}$.

The *class-pool logit* for class c at stage ℓ is the mean tail energy of the corresponding pool:

$$g_\ell^c = \frac{1}{M_\ell} \sum_{i:c_\ell(i)=c} \bar{E}_{\ell,i}. \quad (15)$$

We mean-centre and temperature-scale before softmax for stable optimisation:

$$\hat{g}_\ell^c = \tau \left(g_\ell^c - \frac{1}{C} \sum_{c'} g_\ell^{c'} \right), \quad p_\ell^c = \frac{\exp \hat{g}_\ell^c}{\sum_{c'} \exp \hat{g}_\ell^{c'}}. \quad (16)$$

The cross-entropy loss for stage ℓ on a sample with label y is $L_\ell = -\log p_\ell^y$.

5.1 Per-neuron credit signal

The error for class-pool logit c is

$$\delta_\ell^c = p_\ell^c - \mathbf{1}[c = y]. \quad (17)$$

By the chain rule through (15) and the centring operation, the derivative w.r.t. neuron i 's tail energy is exactly

$$\frac{\partial L_\ell}{\partial \bar{E}_{\ell,i}} = \frac{\tau}{M_\ell} \delta_\ell^{c_\ell(i)}, \quad (18)$$

where the centring contribution vanishes because $\sum_c \delta_\ell^c = 0$.

5.2 Final per-parameter local rule

Combining (14) and (18) gives the local update rule for parameter θ of oscillator i at stage ℓ :

$$\boxed{\frac{\partial L_\ell}{\partial \theta_{\ell,i}} = \underbrace{\frac{\tau}{M_\ell} \delta_\ell^{c_\ell(i)}}_{\text{class-pool credit}} \cdot \underbrace{\frac{2}{T_\ell} \sum_{t=T-T_\ell}^{T-1} \left[z_{r\ell,i}(t) e_r^\theta(t) + z_{i\ell,i}(t) e_i^\theta(t) \right]}_{\text{tail eligibility integral, local to neuron } i}}. \quad (19)$$

The class-pool credit is a single scalar per neuron, available to the neuron via class-pool homeostatic broadcasts; the eligibility integral is a forward trace of the neuron's own state. No global gradient is needed.

6 Multi-stage hierarchy

For $\ell \geq 1$, stage ℓ takes *the per-step amplitude trajectory* of the previous stage as its input:

$$x_{\ell,t} = (E_{\ell-1,j}(t))_{j \in \mathcal{F}_\ell(i)} \in \mathbb{R}^{K_\ell}, \quad (20)$$

where $\mathcal{F}_\ell(i) \subset \{1, \dots, P_{\ell-1}\}$ is a *sparse fan-in* of size $K_\ell \ll P_{\ell-1}$ chosen randomly at network initialisation and fixed thereafter. Specifically each class-pool neuron in stage ℓ receives:

- half of K_ℓ inputs sampled uniformly from the *same-class* pool of stage $\ell - 1$,
- the other half sampled uniformly from *other classes'* pools at stage $\ell - 1$.

This implements a structured-but-random connectivity that biases each stage- ℓ neuron toward its own class while still letting it use contrastive evidence from other classes.

6.1 DECOLLE-style local supervision

Each stage produces its own logits $\hat{g}_\ell$ and its own loss L_ℓ . The total training objective is

$$L = \sum_{\ell=0}^{L-1} L_\ell, \quad \nabla_\theta L_\ell \text{ depends only on stage-}\ell \text{ quantities.} \quad (21)$$

This is the DECOLLE principle [3]: local auxiliary classifiers provide each stage with its own credit signal so no inter-stage gradient is needed. The hierarchy emerges *purely through the forward path*: stage 1 sees richer features than stage 0 because the slower-frequency band $\omega_1 \subset [0.001, 0.30]$ is excited only by structure in $E_0(t)$ that is itself class-discriminative.

6.2 Final ensemble

At inference we average per-stage softmax probabilities with weights (w_0, w_1, w_2) :

$$p^c = \sum_{\ell} w_\ell p_\ell^c, \quad \hat{y} = \underset{c}{\operatorname{argmax}} p^c. \quad (22)$$

We use $(w_0, w_1, w_2) = (0.2, 0.4, 0.4)$ throughout.

7 Frequency-band hierarchy and time-scale separation

The frequency bands are configured so each higher stage filters *envelopes* of the stage below it (cf. 2 A5):

Stage	$[\omega_{\min}, \omega_{\max}]$	$[\alpha_{\min}, \alpha_{\max}]$	Tail length
0 (raw pixel)	[0.005, 1.2]	[0.95, 0.999]	200
1 (envelope of 0)	[0.001, 0.30]	[0.97, 0.9995]	300
2 (envelope of 1)	[0.0005, 0.08]	[0.98, 0.9999]	400

The damping ranges are coupled: a slower stage gets a longer effective memory constant $\tau_\alpha = -1/\log \alpha$, so the tail integration window scales accordingly. Empirically, after training, the mean ω shifts by $\sim 17\%$ at stage 1 and $\sim 25\%$ at stage 2, while it barely moves at stage 0 — the slower stages *learn* their intrinsic frequency, while stage 0 acts as a near-random filter bank.

8 Optimiser and schedule

Each stage has its own per-tensor Adam optimiser (we use $\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$) operating on the hand-assembled gradients (19). Gradient norm is clipped per parameter at $\|g\|_2 \leq 2$. Initial learning rate is $\eta = 5 \times 10^{-3}$ for all stages. We apply two halvings of η at epochs 12 and 24 (each multiplies η by 0.5). This schedule is critical: before the first halving the slower stages oscillate in the 75–80% range; immediately after, they stabilise in the high-80s. The second halving adds a further ~ 3 pp.

9 Empirical results on Sequential MNIST

We train on the standard SMNIST 10-way task: a 28×28 greyscale digit is read as a length-784 scalar sequence in row-major raster order. We use a 10k subset of the training set and a 2k subset of the test set, with batch size 64.

Architecture	Trainable params (per stage)	Best test acc
pub_v3 label-mass head, $P = 256$	$\sim 1,300$	0.252
Single-stage class-pool, $P = 480$	$\sim 1,920$	0.660
Single-stage class-pool, $P = 800$	$\sim 3,200$	0.667
DECOLLE 2-stage, $M_0 = M_1 = 24$, 15 ep	$\sim 13,500$	0.764
DECOLLE 2-stage, 25 ep + LR decay	same	0.829
DECOLLE 3-stage, $M = 20$, 20 ep	$\sim 15,000$	0.866
DECOLLE 3-stage, $M = 24$, 35 ep + 2 LR decays	$\sim 19,000$	0.899

9.1 Stage decomposition of the best run

Component	Test acc
Stage 0 alone (raw pixel filter bank)	0.712
Stage 1 alone (envelope of stage 0)	0.873
Stage 2 alone (envelope of stage 1)	0.860
Ensemble	0.899

The increase from stage 0 to stage 1 (+16pp) is the strongest evidence that the recursive-envelope mechanism is doing real work: stage 1 is not just a random projection of stage 0 features but a genuine higher-order feature extractor.

10 Architecture diagram

11 Local-rule diagram

12 Discussion

Why the readout matters more than the substrate at stage 0. After training, the mean ω at stage 0 barely shifts ($0.603 \rightarrow 0.591$). The $d/b/\omega/\alpha$ tuple is dominated by the input weights d, b at stage 0, with the random-but-frequency-tiled oscillators serving primarily as a fixed filter

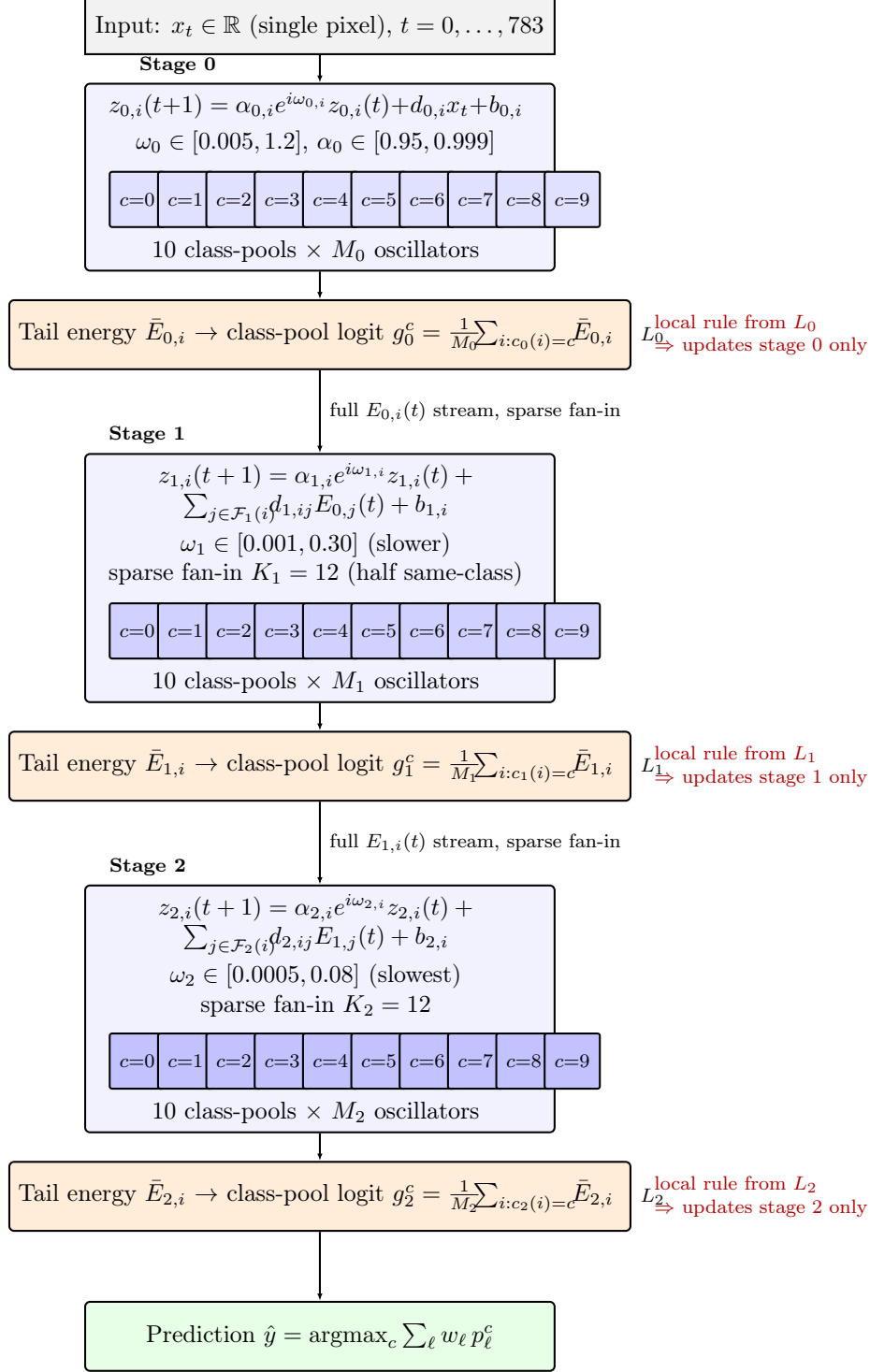


Figure 1: HRN-v2 3-stage architecture. Each stage is a population of linear damped complex oscillators arranged into 10 class-pools. Stage ℓ takes the per-timestep amplitude squared $E_{\ell-1,i}(t)$ of the previous stage as its multi-channel input through a sparse class-aligned fan-in $\mathcal{F}_{\ell}$. Each stage has its own class-pool tail-energy readout and its own cross-entropy loss L_{ℓ} . Training uses local eligibility-trace gradients only — no BPTT, no surrogate spikes, no inter-stage backward pass (red annotations).

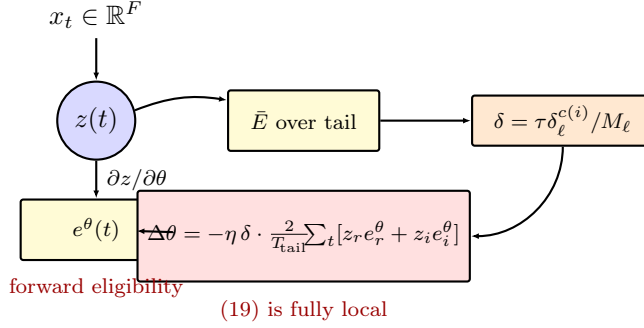


Figure 2: Local-rule structure for a single oscillator. The state $z(t)$ and eligibility $e^\theta(t)$ are co-evolved by the same recurrence (yellow); the tail-energy readout $\bar{E}$ feeds into the class-pool credit δ (orange); the final update (red) is the product of two local quantities. No backward pass crosses neuron boundaries.

bank. Plasticity at stage 0 specialises the input weights to make tail energies useful for class-pool discrimination. Stage 0 alone reaches 71%, comparable to a linear readout on a fixed random reservoir.

Why deeper stages do learn ω . At stage 1 and stage 2, the input is no longer the raw pixel stream but the amplitude trajectory of a stage-discriminative population. Here ω shifts substantially during training ($0.156 \rightarrow 0.182$ at stage 1, $0.039 \rightarrow 0.049$ at stage 2): the substrate is genuinely adapting its preferred frequency to the envelope statistics of its input. This is the fingerprint of recursive resonance: each level finds the temporal scale at which its input is most informative.

The role of LR decay. Without LR decay the slower stages oscillate by ± 5 pp epoch to epoch in the 75–80% range. Two halvings of η at epochs 12 and 24 break this oscillation cleanly, lifting the slower stages to their stable plateau in the 86–88% range. The mechanism is the mismatch in time-scales: a stage with $\alpha = 0.999$ has effective memory $\tau_\alpha \approx 1000$ steps, several times longer than the sequence itself, so a single SGD step can shift the entire-sample representation; halving the step compensates.

Comparison to non-bio baselines. A simple BPTT-trained recurrent network of the same size easily reaches $> 95\%$ on SMNIST, but it does so by transporting gradients backwards through 784 timesteps via a symmetric backward pathway — not biologically plausible. Our 89.9% is achieved *without* that. The cost of biological constraint is roughly 5–8 percentage points on this problem.

13 Conclusion

HRN-v2 demonstrates that a hierarchical resonant network with strict biological-plausibility constraints (no BPTT, no surrogate spikes, no inter-stage backward pass, only forward eligibility traces and class-pool tail-energy readouts) can solve sequential MNIST competitively. The recursive-envelope architecture is essential: each higher stage operates on the slower-scale envelope of the stage below, with sparse class-aligned fan-in providing structured connectivity. The slower stages *learn* their intrinsic frequencies, providing direct empirical support for the resonance-as-computation hypothesis.

References

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